

Multiple Choice (10 marks)

1. The amino acid ornithine is regenerated during each turn of the urea cycle in a similar manner to which of the following participants in the citric acid cycle?
 - (a) Alpha-ketoglutarate
 - (b) Glutamate
 - (c) Acetyl CoA
 - (d) Oxaloacetate
 - (e) Citrate
2. In plants, which tissue is responsible for new growth?
 - (a) Apical tissue
 - (b) Sclerenchyma tissue
 - (c) Ground tissue
 - (d) Vascular tissue
 - (e) Cork tissue
3. A feature of amino acids NOT found in carbohydrates is the presence of
 - (a) carbon atoms
 - (b) oxygen atoms
 - (c) nitrogen atoms
 - (d) hydrogen atoms
4. What are some of the cues used by animal in migrating?
 - (a) Animals use the sound of the ocean waves for directional cues in migration.
 - (b) Animals rely solely on memory from previous migrations.
 - (c) Animals migrate based on the patterns of the stars alone.
 - (d) Animals follow changes in air pressure for migration.
 - (e) Animals use temperature changes for migration.
5. All of the following statements about plant embryogenesis are correct EXCEPT:
 - (a) The suspensor is derived from the basal cell.
 - (b) Cotyledons are derived from the apical cell.
 - (c) Shoot apical meristem formation occurs after seed formation.
 - (d) Precursors of all three plant tissue systems are formed during embryogenesis.

True / False (10 marks)

Answer True or False

6. True or False: The female bacterium becomes a male due to nutrient deficiency, which alters its genetic composition.
7. True or False: Arthropods use their wings to facilitate movement in various environments.
8. True or False: Genetic variation must be present in a population before natural selection can act upon that population.
9. True or False: Anesthetics stimulate the release of endorphins to mask pain, thereby enhancing the body's natural pain relief mechanisms.
10. True or False: Removing broken eggshells provides extra nutrition to the parents, who consume the shells after the chicks hatch.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Discuss the hormonal changes that occur during parturition, focusing on the roles of estrogen, progesterone, oxytocin, relaxin, and prostaglandins in the delivery process.
12. Discuss the mechanisms by which higher plants eliminate waste products, including diffusion and leaf shedding, and analyze their significance for plant health and ecosystem balance.

End of Paper